Advanced Simulation (2DI66)

Assignment 2: Optimizing waste recycling points in Eindhoven

Waste recycling point: Lodewijkstraat Eindhoven (2012)

Legend

- et Entrance
- ex Exit
- Hall
- Q Hall queue
- H_et Hall entrance
- H_ex Hall exit
- Traffic
- OF Overflow containers

Index

Wc Wood C (Hout C)
F Flammable (Branbaar)
W Wood A&B (Hout A&B)
M Metal (IJzer en metalen)
R Roofing (Dakleer)
Ma Mattresses (Matrassen)
Ca Carpet (Tapijt)
Gp Gypsum (Gips)
E Electronics (AEEA)
P Paper (Papier)
Sl Soil (Grond)
G Green (Blad en gras)
A Asbestos (Asbesthoudend materiaal)
Ch Chemical (Klein Gevaarlijk Afval, KGA)
Sp Soft plastic (plastiek)
Cl Clothes (Kleding)
B Bottles (Flessenglas)
T Tires (Banden)
Dc Debris, clean (Puin, schoon)
Dd Debris, dirty (Puin, vuil)

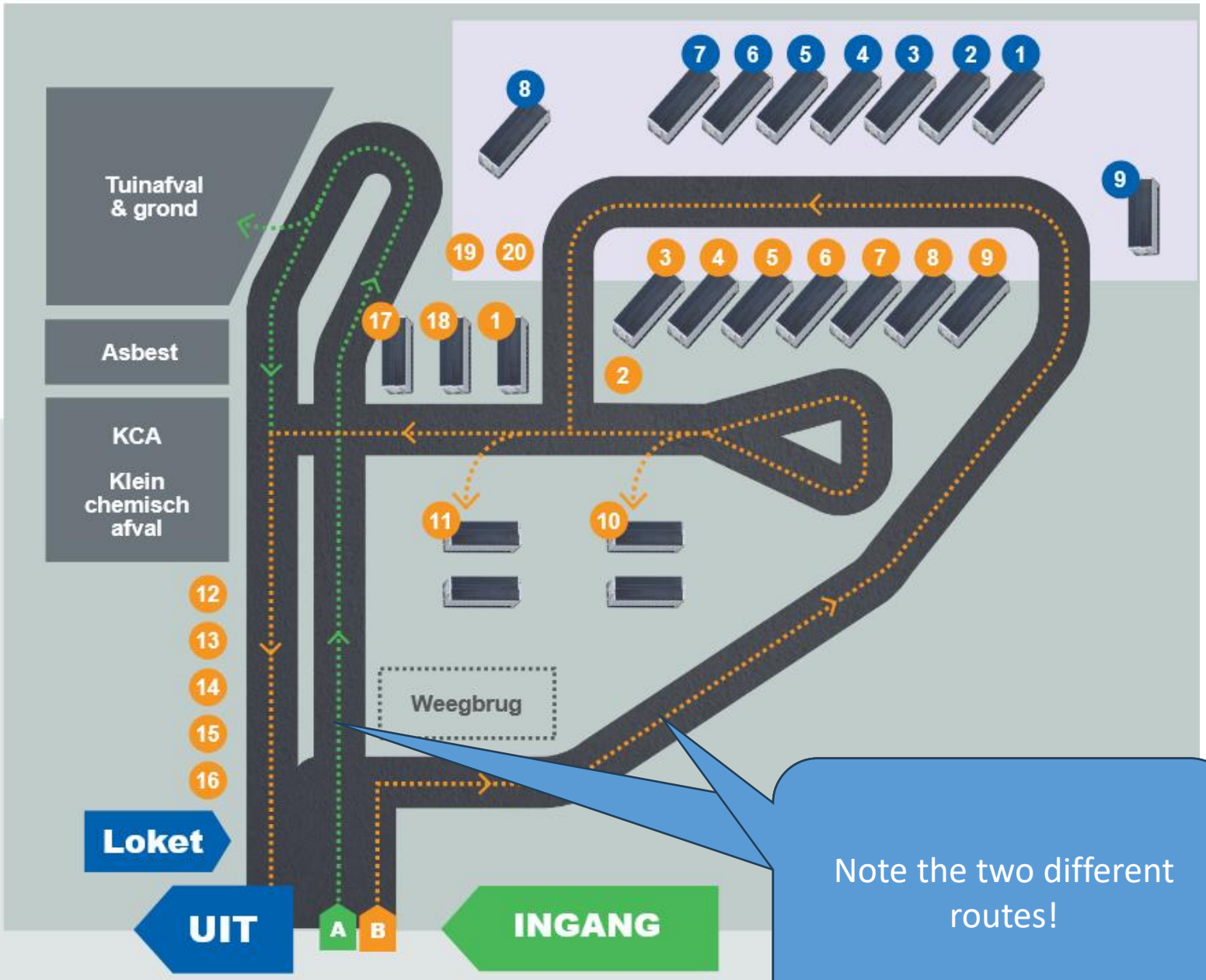
Waste recycling point Lodewijkstraat (2025)

Inside the hall:

1 Wood B
2 Metal
3 Flammable
4 Furniture: Couches
5 Wood B
6 Flammable
7 Carpet
8 Flammable
9 Paper

Outside the hall:

1 Wood C
2 Flammable
3 Glass
4 Flammable
5 Paper
6 Metal
7 Flammable
8 Wood B
9 Garden furniture
10 Debris Dirty
11 Debris Clean
12 Tires
13 Glass bottles
14 Plastics
15 Textile
16 Tires
17 Roofing
18 Gypsum
19 Mattresses
20 Electronics



Note the two different routes!

Entrance

Queue outside the Hall

Inside the Hall (1 – 9)

Pictures from 2012. This looks slightly different today.

“Overflow” outside the Hall (1 – 9 and 17 – 20)

→ Only for small cars

Debris clean + dirty (DcDd) (10 + 11)

Pictures from 2012. This looks slightly different today.

Rest (12 – 16)

Picture from 2012. This looks slightly different today.

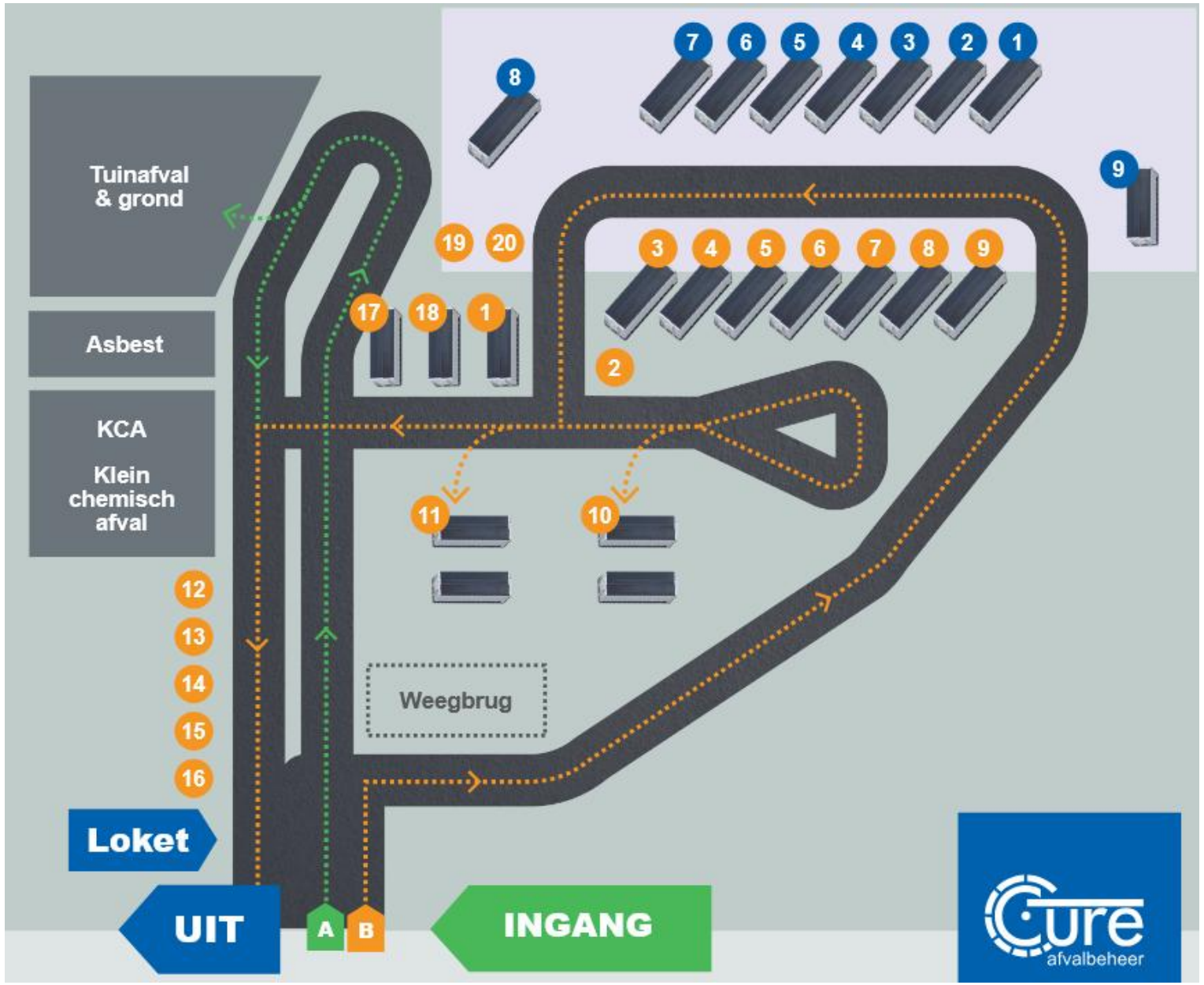
The Simulation Model

Stations and parking places:

- Entrance (1)
- Hall (12)
- Overflow (10)
- Debris Clean/Dirty (7)
- Green (5)
- Rest (5)

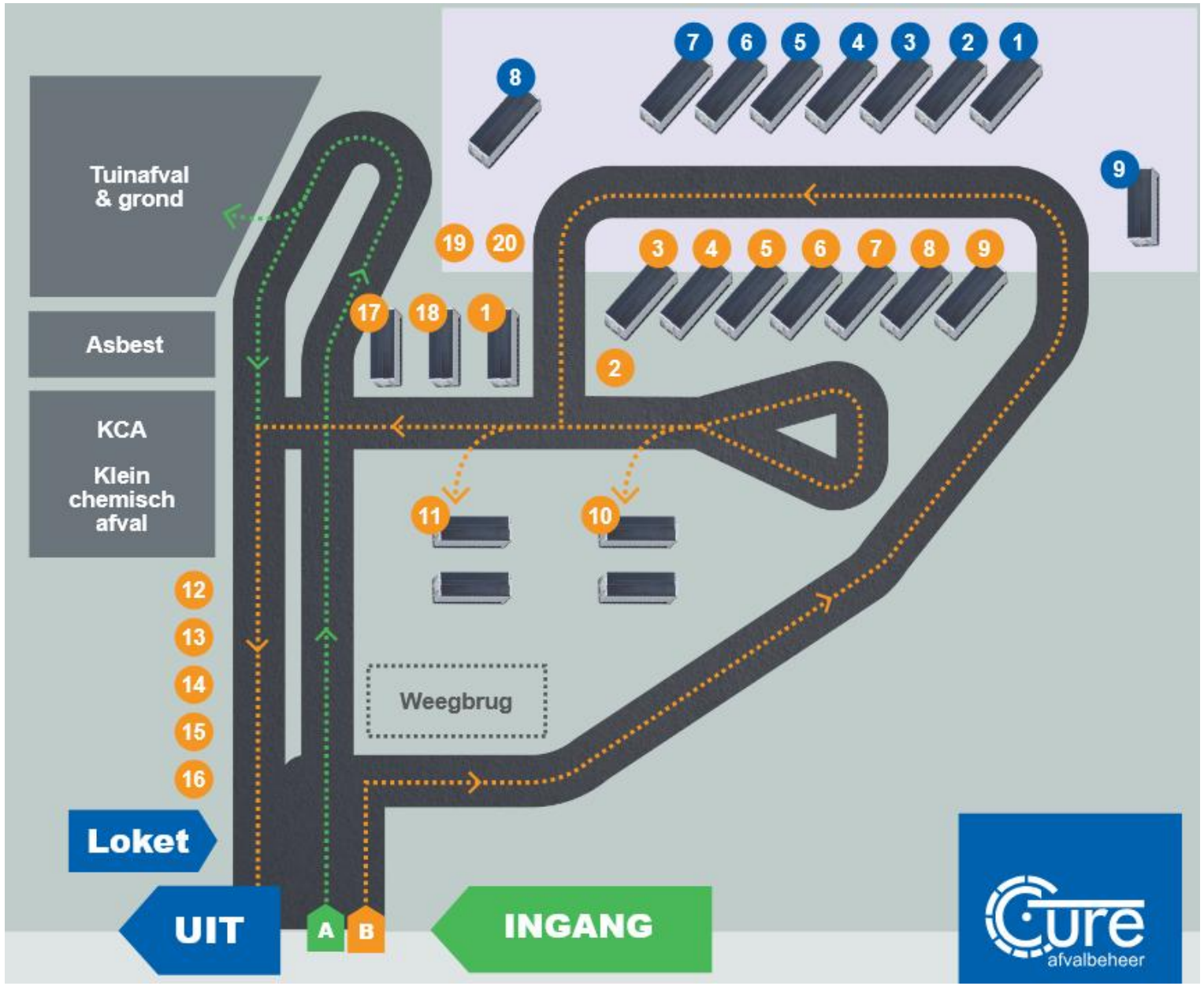
Big cars (van or car+trailer):
require 2 parking spaces!

Routing and Blocking:



- Route A for cars that only visit “green” and/or “rest”
- Route B for cars visiting “hall”, “overflow”, and/or “DcDd”
- Combination Hall/Overflow/DcDd + Green is NOT allowed!

Routing and Blocking:

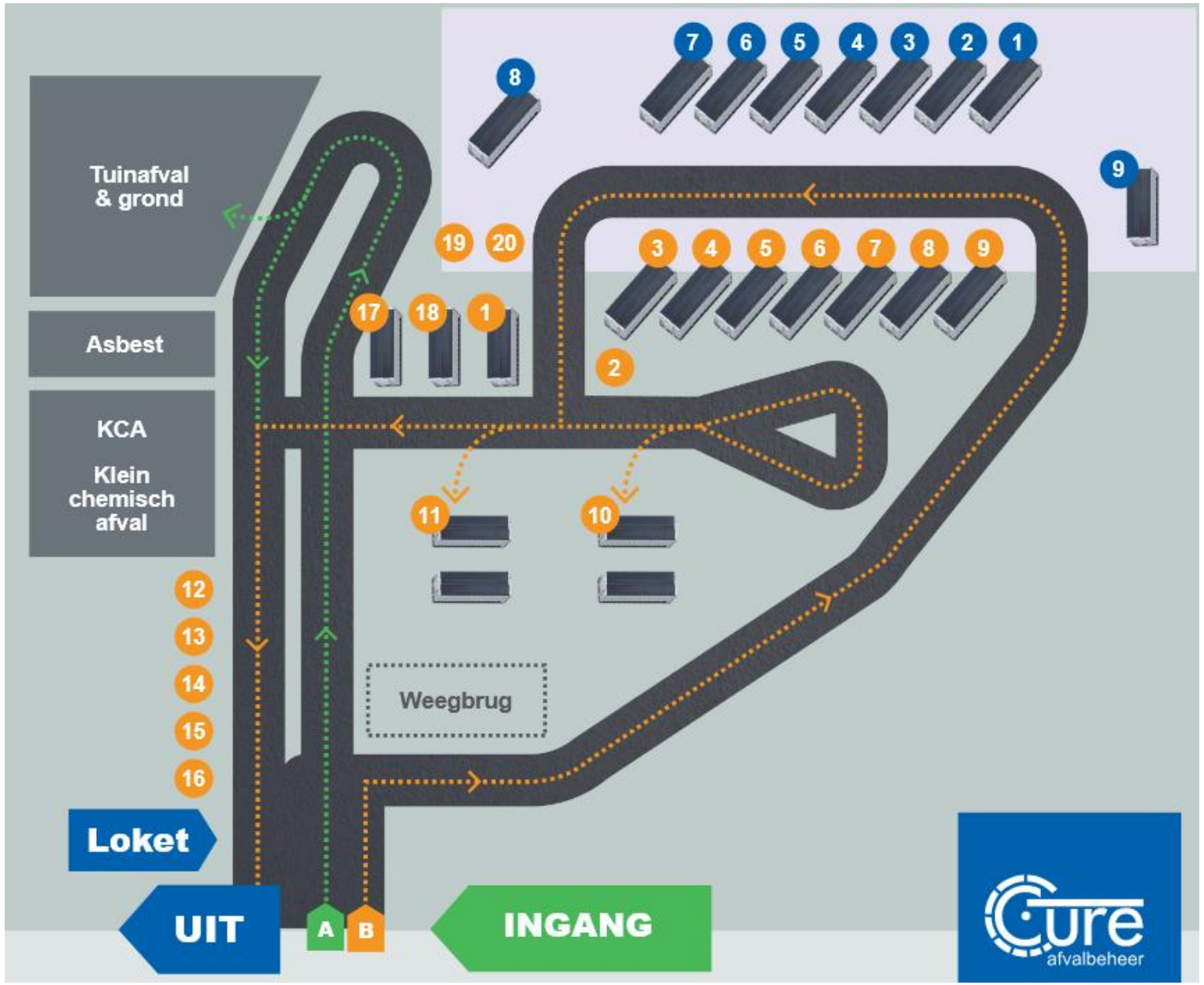


- Route A for cars that only visit “green” and/or “rest”
- Route B for cars visiting “hall”, “overflow”, “DcDd”, and/or “rest”
- Combination Hall/Overflow/DcDd + Green is NOT allowed!

For customers who need to visit hall or overflow:

- Overflow only allowed for small cars
- Big cars all go to the hall
- Only when overflow is full, small cars are allowed inside Hall
- When hall is full, the next big car will be blocked in front of the hall and starts a queue
- Small cars cannot skip the queue

Routing and Blocking:

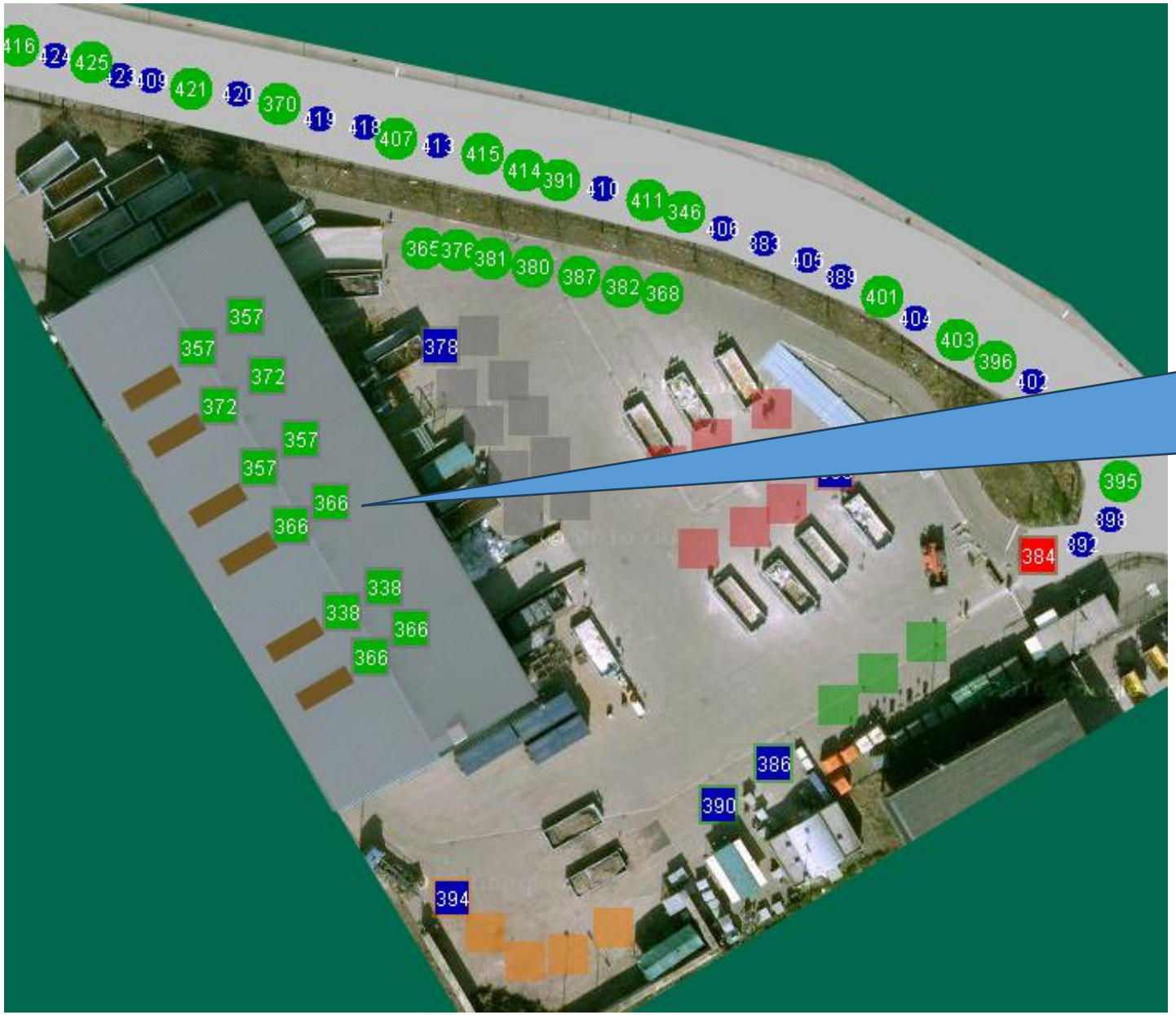


- A car can only move to the next station if there is a sufficiently large available parking spot there.
- As long as there is no available parking spot, the car stays in its place, occupying the current parking spot

Example:

- A large car is in the hall, finished service. Wants to go to DcDd
- There is only 1 available parking spot at DcDd
  →
  has to wait in the hall for another car to leave at DcDd and keeps that parking spot occupied

Routing and Blocking Example



6 big cars in Hall → Hall is full

Routing and Blocking Example

Car in front of queue needs to be in the hall, but cannot proceed because Hall is full.
So it blocks the cars behind

Routing and Blocking Example

Queue in front of Hall = 7 big cars long
→
Car at entrance wanting to go to DcDd is blocked!

Data (Cleaned)

- Opening hours
- Arrival rates throughout the day
- Percentage big cars
- Service times in all stations
- Number of parking spaces in each station
- Maximum queue length Hall
- Customer classes and routing

Data (Cleaned): Customer classes and routing

Customer class:	Hall, Debris, Green, Rest 2						
Percentage of total:	7%						
Routing matrix				**To**			
			Hall/Overflow	**Debris (clean + dirty)**	**Green**	**Rest**	**Exit**
		Entrance	100%	0%	0%	0%	0%
		Hall/Overflow		37%	12%	6%	45%
	From	**Debris (clean + dirty)**			33%	0%	67%
		Green				0%	100%
		Rest					100%

Careful: this data is for the old situation, when routing still allowed to go from Hall/Overflow/DcDd to Green

Main questions

- Can you simulate the current layout of the Lodewijkstraat WRP, using the data from 2012 as input?
- The municipality of Eindhoven wants to close one of the other WRPs, in Esperheide, which will lead up to 20% more customers at the Lodewijkstraat WRP. How will this affect waiting times and queue lengths?
- Can you suggest improvements that significantly improve this performance? Think of: shortening service times, adding parking spots, changing the routing, creating different stations.

Your final grade

In general, a more realistic model will also result in a higher grade (as long as the results are correct obviously). You can include:

- Reneging from the queue
- Different service times for big/small vehicles
- Probability of customer forgetting city pass
- Including bicycles and people on foot, who get priority

A final note

- Although you are allowed to use AI tools to assist you with this assignment, we stress (again) that you should always honestly cite your sources including an indication of where and how you used AI
- You are not allowed to use any packages that have not been used during the lectures (or at least ask for permission)
- Your code should really be based on the sample code provided during the lectures. In particular, classes like the future event set, events, customers, simulation results, and the structure of the main simulation should have a similar setup.

Good luck!